

Activity

Core temperature and post-surgery infections

When patients undergo surgery, the anesthesia received can lower their core temperature. While surgical environments are controlled and aim to regulate patient core temperature, some degree of cooling can still occur. *Hypothermia* during surgery is said to occur if the core temperature is below 36°C, and we might suspect that greater differences from 36°C would be associated with worse outcomes. Furthermore, it is possible that there is a relationship between core temperature and patient outcomes even within the normal range of body temperatures (36 – 37.5°C).

Walters *et al.* (2020) conducted a study to investigate the relationship between core temperature and the risk of serious infection in colorectal surgery patients. Their study included 7908 adult patients who received colorectal surgery between January 2005 and December 2014, and who had general anesthesia for a surgery lasting at least one hour that included at least 30 minutes of esophageal core temperature monitoring (see Walters *et al.* for the full inclusion/exclusion criteria). The study authors then recorded whether each patient developed a serious infection within 30 days after surgery.

Research question: What is the relationship between core temperature during surgery, and the risk of developing a serious infection after surgery?

Your task

You have been approached by the researchers who conducted this study. They want you to help them plan an analysis, using a regression model to answer their question of interest.

Last time we worked with this data, we decided to include the following explanatory variables in the model, with Serious Infection as the response:

- TWATemp (measure of core temperature during surgery)
- Surgery information – duration (SurgDuration) and open vs. laproscopic (Open)
- Patient characteristics (Age, BMI, FEMALE)
- Information on pre-existing health conditions and comorbidities (CharlsonScore, CHF, VALVE, ..., ANDEMDEF)
- Drug information (DRUG, SteroidHx, ImmunosuppressantHx)

Questions

1. What assumptions is your model making?

- shape: log-odds has a linear relationship w/ explanatory variables
- Independent observations
- Randomness (data come from some kind of random process)

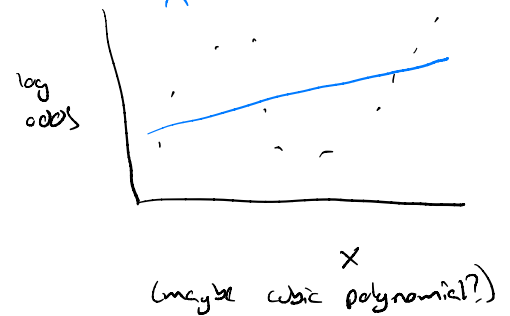
2. What model diagnostics will you perform to assess whether the model is a reasonable fit to the data? Describe any plots you would make and statistics you would calculate, and how you would use the results – what are you looking for? What would the diagnostics look like if the model is a good fit? What would they look like if the model is *not* a good fit?

· Shape: empirical logit plot

x axis: explanatory variable

y axis: empirical log odds

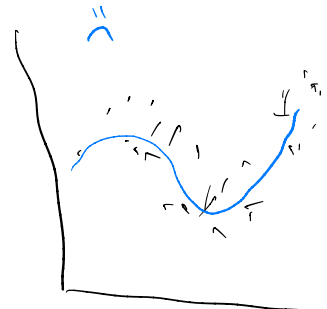
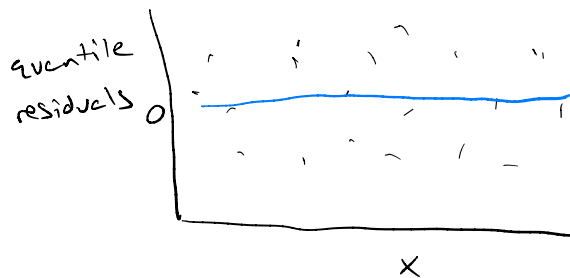
(quantitative – a categorical variable doesn't have the same sense of linearity)



(randomized) quantile residual plot

x axis: either fitted values, or explanatory variable

y axis: quantile residuals



Questions

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Multicollinearity: - pairs plot (plot scatterplots of explanatory variables against each other, correlations between explanatory variables)

Concerned if there are variables with very high correlation

- variance inflation factors (VIFs)
Concerned about high multicollinearity
(common thresholds include $VIF > 5$
or $VIF > 10$)

Outliers:

- look at unusual observations in EDA
(e.g. values far in the tails of distribution on a boxplot or histogram)

- use Cook's distance to check how much estimated coefficients change with and without potential outliers
Typical thresholds for Cook's distance include $2 \times 0.5 \{ 1$ (e.g. concerned about values > 1)

3. If model diagnostics suggest the model is *not* a good fit, what modifications would you explore to improve model fit?

- Issues with shape assumption: explore transformations of explanatory variables (log, square root, polynomial, etc.)
- Multicollinearity:
 - If primary variable of interest (e.g. TWATemp) is not affected (does not have high VIF), maybe not concerned
 - Otherwise, consider using a subset of affected explanatory variables, or combine several variables
- outliers: report main conclusions (effect of TWATemp) both with and without potential outliers

4. Once you have a final model with which you are satisfied (model diagnostics look sufficiently reasonable), how will you use that final model to address the research question of interest?

compare models with and without TWATemp

• E.g., if our model looks like

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \text{TWATemp}_i + \dots$$

Then test $H_0: \beta_1 = 0$ $H_A: \beta_1 \neq 0$

or if our model looks like

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \text{TWATemp}_i + \beta_2 \text{TWATemp}_i^2 + \dots$$

$$H_0: \beta_1 = \beta_2 = 0$$

H_A : at least one of $\beta_1, \beta_2 \neq 0$